

MALE UAV Pipeline Test Report

End-to-end data flow, model behavior, controls, and dashboard

OVERALL RESULT

Pipeline connectivity passed; health assessment and dashboard data integrity have material faults.

Test date: 25 September 2026 **Scope:** Layers 1–7, Redis streams, model inputs/outputs, manual controls, and dashboard adapter.

Retest conclusion. The reported data-path and dashboard-integrity faults have been corrected in code: dashboard sensors now come only from valid, fresh Layer 2 frames; controls replace previous override flags and carry revisions/provenance through Layers 2–5 and the assessment; profile settings preserve requested values; unavailable residual/model/RUL values are no longer replaced with nominal-looking defaults. Layer 6 now blocks the bundled synthetic artifact and windows with insufficient per-channel coverage. The current classifier and RUL outputs remain unavailable for decision use until verified data and validated priors are supplied.

Test execution and coverage

- Automated suite after fixes: 69 tests passed; 1 optional legacy-dataset test skipped. Redis-backed tests ran against a temporary local Redis server, which was stopped after the suite.
- Retest coverage: unit and Redis integration checks cover Layers 1–7 contracts, synchronized frame validity and staleness, timestamp/provenance propagation, fixed-window availability, classifier gates, Layer 7 publication, and dashboard adaptation. Compilation passed for app, layers 1–7, dashboard, and tests.
- Current live visual dashboard retest was not repeated. The prior Streamlit AppTest was inconclusive because automatic refresh prevented completion. No claim of full visual UI verification is made.
- Control-path retest: code now increments and returns control_revision, replaces active override flags on each control request, passes provenance through synchronization and feature windows, and marks the dashboard assessment pending until revisions match. Focused automated tests verify the Layer 2 → Layer 3 → Layer 4 → Layer 5 context path.
- Manual UI interaction and browser rendering remain unverified after these code changes; the API/control contracts and dashboard payload formatting were checked with unit tests and compilation.

Findings

Severity	Retest status	Evidence and remaining dependency
Critical	Mitigated	The synthetic, unverified Layer 6 artifact is gated. No classifier prediction is emitted from it. Verified model data is still required for decision use.
High	Mitigated	Per-channel residual availability below 0.05 blocks classification. Simulator cadence is 2 Hz; held points on the 100 Hz grid are not new observations.
High	Dependency	Fuel-flow prediction needs FUEL_DENSITY_KG_L. Without it, flow residual coverage is zero and classifier inference is gated.
High	Surfaced	PINN out-of-domain and missing-input outputs remain unavailable. Model domains need calibration against the intended operating envelope.
High	Mitigated	Control revisions and simulated provenance pass through frames/windows. Assessment pending state compares revisions; focused tests cover this path.
Medium	Mitigated	Dashboard joins the latest measurement-updated EKF result and its timestamp. Raw sensors have no simulator fallback.

Medium	Mitigated	Atmosphere API preserves requested altitude and temperature. ISA_STANDARD is the initial UI selection.
Medium	Mitigated	Each manual control request replaces the active override flags; omitted controls are cleared.
Medium	Mitigated	Risk-derived health is no longer labeled EHI. Unvalidated RUL is shown as unavailable.

Data contract and architecture notes

- Dashboard sensor contract after fixes: raw values are sourced exclusively from a fresh, active-session Layer 2 frame, either directly from engine:synced:frames or from a valid Layer 2 frame embedded in the assessment. Missing, stale, or invalid channels display as unavailable. PINN predictions and EKF residuals remain separate values.
- Layer 5 emitted 672 named features, matching the loaded LightGBM artifact. Isolation Forest intentionally uses a 222-feature subset. The repository report and v4 registry show 672 features; a separate integration note claiming 690 is stale.
- Layer 7 risk remains advisory. Numeric RUL is hidden unless its prior is validated for decisions. The checked-in Layer 6 model is a synthetic development artifact; the service now withholds its classification. No real aircraft data, maintenance outcomes, or verified healthy sorties were introduced by these fixes.

Corrections applied

- Dashboard sensor values use Layer 2 only and invalid/stale values render as unavailable; fabricated sensor, health, and RUL defaults were removed.
- Control revision and simulated-input provenance now travel with the synchronized frame and through the physics, EKF, window, classifier assessment, and dashboard joins. Override selection is replaced per request.
- Classifier inference now fails closed for the current unverified artifact and when any residual channel has less than 0.05 availability. PINN out-of-domain outputs remain explicitly unavailable; fuel-flow physics still requires FUEL_DENSITY_KG_L.
- The dashboard distinguishes risk-derived health from an independent health index and withholds unvalidated RUL values. Model accuracy and RUL validity still require independently verified operational data and domain evaluation.
- The dashboard takes its residuals from the latest valid measurement-updated EKF record and carries the residual timestamp. Per-channel raw sensor age/source/rate/validity remain exposed.

Limitations

Limits after retest: tests used the simulator, bundled synthetic artifacts, and local Redis. They validate code-path behavior and data contracts, not aircraft diagnostic accuracy or operational suitability. The simulator emits at 2 Hz while Layer 2 uses a 100 Hz grid with held values between fresh samples. Fuel-flow predictions need configured density. The current PINN training domains can still reject simulator states. These data/model limitations require calibrated artifacts and representative, independently verified engine data; code changes cannot establish those facts.